

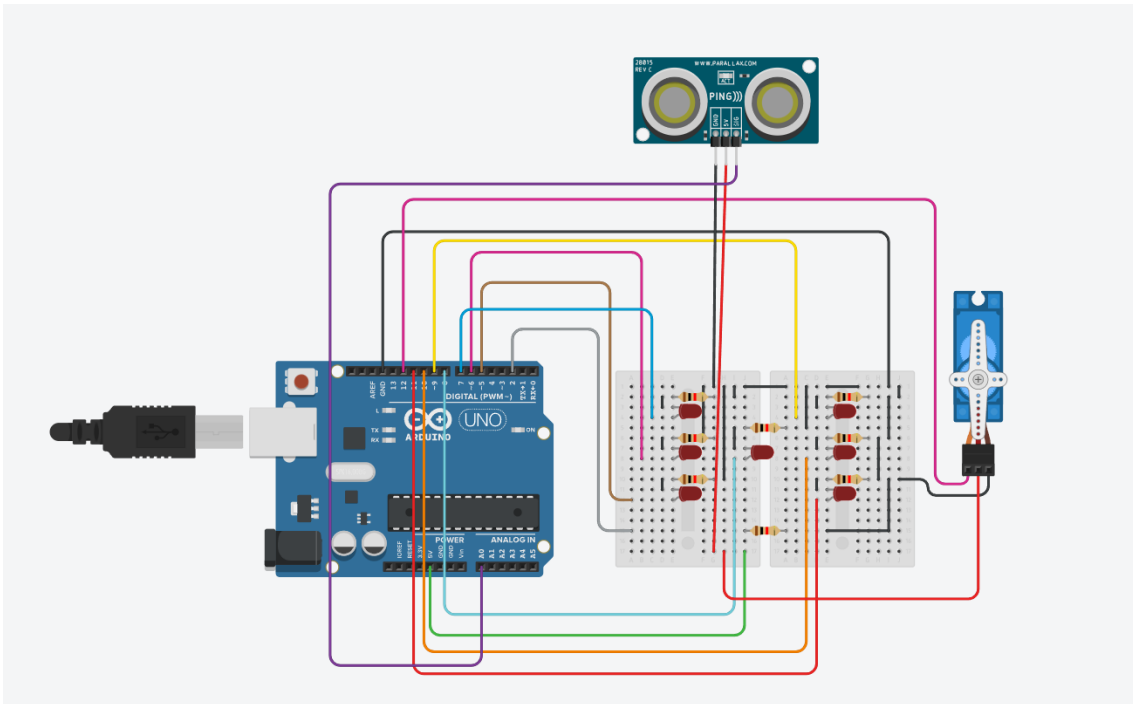
ATIVIDADE:

SISTEMAS EMBARCADOS E IOT

NOMES:

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PRINT E CÓDIGO:



```
#include <Servo.h>
```

```
Servo servo1;
```

```
const int PINO_SENSOR = A0;
```

```
const int PINO_SERVO = 12;
```

```
const int NUM_LEDS = 6;
```

```
const int LEDS[NUM_LEDS] = {3, 4, 5, 6, 7, 8};
```

```
long readUltrasonicDistance(int triggerPin, int echoPin) {
```

```
    pinMode(triggerPin, OUTPUT);
```

```
    digitalWrite(triggerPin, LOW);
```

```
    delayMicroseconds(2);  
    digitalWrite(triggerPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(triggerPin, LOW);  
    pinMode(echoPin, INPUT);  
    return pulseIn(echoPin, HIGH);  
}
```

```
void setup() {  
    for (int i = 0; i < NUM_LEDS; i++) {  
        pinMode(LEDS[i], OUTPUT);  
    }
```

```
  
    servo1.attach(PINO_SERVO);  
    servo1.write(0);  
    randomSeed(analogRead(A1));  
    Serial.begin(9600);  
}
```

```
void loop() {  
    float centimetro = 0.01723 * readUltrasonicDistance(PINO_SENSOR,  
PINO_SENSOR);  
  
    Serial.print("Distancia: ");  
    Serial.print(centimetro);  
    Serial.println(" cm");
```

```
if (centimetro < 40) {  
    int dado = random(1, 7);  
    Serial.print("Dado sorteado: ");  
    Serial.println(dado);  
    for (int i = 0; i < NUM_LEDS; i++) {  
        digitalWrite(LEDs[i], i < dado ? HIGH : LOW);  
    }  
    if (dado % 2 == 0) {  
        Serial.println("PAR! Liberando premio.");  
        servo1.write(90);  
        delay(1500);  
        servo1.write(0);  
    } else {  
        Serial.println("IMPAR. Sem premio.");  
    }  
    delay(3000);  
    for (int i = 0; i < NUM_LEDS; i++) {  
        digitalWrite(LEDs[i], LOW);  
    }  
}  
  
delay(200);  
}
```